

FUZE

RISK MANAGEMENT PLAN

Version 1.2
7 Oct 2026

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VERSION HISTORY

Faye Chan, Jethani Samarth and Sinha Armaan lead the preparation of the Risk Management Plan. Mohika Kataruka maintains the controlled version and records agreed revisions.

Version #	Implemented By	Revision Date	Approved By	Approval Date	Reason
1.0	Jethani Samarth and Sinha Armaan	25 Sep 2026	Mohika Kataruka	28 Sep 2026	Initial Risk Management Plan draft
1.1	Faye Chan	3 Oct 2026	Mohika Kataruka	5 Oct 2026	Revised document
1.2	Faye Chan	7 Oct 2026	Mohika Kataruka		Aligned with Project Plan 1.1: team likelihood estimates, effort figures, contract date and tools. Rated risks 2, 3 and 5; added risks 14–17; updated risk 9 status.

UP Template Version: 11/30/06

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1 INTRODUCTION

1.1 PURPOSE OF THE RISK MANAGEMENT PLAN

A risk is an event or condition that, if it occurs, could have a positive or negative effect on a project's objectives. This Risk Management Plan defines how risks associated with the FUZE project will be identified, analyzed, managed and reported. The plan supports the project's incremental development from 20th August to 25th November 2026. It covers development, integration, testing, deployment and final delivery. Its starting risks and estimates are taken from Section 8 of the FUZE Project Plan while requirements and quality controls are drawn from the SRS and Quality Plan.
The intended audience is Team FUZE and Ajith Santhisenan, the project sponsor and management.

2 RISK MANAGEMENT PROCEDURE

2.1 PROCESS

Mohika Kataruka, Project Manager, will serve as Risk Manager. Jethani Samarth and Faye Chan lead preparation and coordination of the plan. The team will identify risks, assess probability and impact, assign actions to the relevant work-package owners, apply the selected response and review the outcome at weekly meetings and milestone reviews. Scope changes require the sponsor's written approval. (Project Plan, Sections 2, 8 and 10.) Risks will be identified as early as possible in the project so as to minimize their impact. The steps for accomplishing this are outlined in the following sections.

2.2 RISK IDENTIFICATION

Risk identification will involve the project team and sponsor and will review the project scope, deliverables, assumptions, external dependencies, work breakdown structure, effort estimates and quality targets. The 17th September meeting specifically calls for external API failure and rate limits, missing component data, model/API delays, SSE failures, scope changes and schedule delay to be addressed. The Project Plan also identifies underestimated effort, hosting failure, insufficient usability testers and insufficient sample runs.
The Risk Management Log will be maintained with the project's GitLab repository and Trello board with large supporting evidence files in the team's shared drive. These are the project record locations identified in the Project Plan. Repository: https://10.96.189.23/sc3040/TEL1/b (reachable on the NTU network or VPN).

2.3 RISK ANALYSIS

Each identified risk will be assessed for its effect on project cost, schedule and performance. Qualitative analysis will determine the response priority. The initial percentages and impact ratings are the team's estimates and will be reviewed weekly rather than treated as measured occurrence rates.

2.3.1 Qualitative Risk Analysis

The probability and impact of occurrence for each identified risk will be assessed by the project manager, with input from the project team using the following approach:

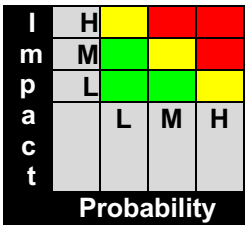
Probability

- High – Greater than 70% probability of occurrence

- Medium – Between 30% and 70% probability of occurrence
- Low – Below 30% probability of occurrence

Impact

- High – Risk that has the potential to greatly impact project cost, project schedule or performance
- Medium – Risk that has the potential to slightly impact project cost, project schedule or performance
- Low – Risk that has relatively little impact on cost, schedule or performance



Risks in the RED and YELLOW zones will have response planning including mitigation and where documented, contingency actions. GREEN risks remain on the review list. Applying the matrix gives RED priority to distributor failure, missing part data, event stream failure, schedule delay and knowledge concentration. Scope changes, slow model responses, underestimated effort, hosting failure, normalisation drift, illegal or looping repairs, model provider outage and team availability fall in YELLOW. Insufficient usability testers, insufficient sample runs, stale cache interpretation and repository access fall in GREEN.

2.3.2 Quantitative Risk Analysis

The team's likelihood estimates are: distributor service failure 40%; missing part data 70%; slow model responses 40%; event stream failure 30%; scope changes 25%; schedule delay 35%; underestimated effort 40%; hosting failure 30%; insufficient usability testers 25%; insufficient sample runs 10% (reduced from 30% now that the seeded corpus exists); normalisation drift 20%; illegal or looping repairs 20%; stale cache interpretation 15%; knowledge concentration 40%; repository access 50%; model provider outage or credit exhaustion 25%; and team availability 40%. These are initial team estimates, not statistically validated forecasts.

The Project Plan's function point estimate is 1,248.2 person-hours, against the 1,106-hour role budget, a difference of 142.2 hours (12.9%). The plan includes 35.7 hours of contingency within the total estimate. (Project Plan, Section 5.2.)

Schedule exposure is also explicit: deployment on 11 November leaves only one week before M5 on 18 November, while a two-week uptime period is required. The prototype is already deployed, so the observation period will start by 21 October. The supplied documents do not provide risk-specific cost or delay distributions, so expected monetary losses and residual probabilities are not calculated here. (Project Plan, Section 8.)

2.4 RISK RESPONSE PLANNING

Each major risk (those falling in the Red & Yellow zones) will be assigned to a project team member for monitoring purposes to ensure that the risk will not "fall through the cracks". The responsibilities below follow the role assignments in the Project Plan. Mohika retains overall responsibility for the risk register and sponsor escalation.

For each major risk, one of the following approaches will be selected to address it:

- **Avoid** – eliminate the threat by eliminating the cause
- **Mitigate** – Identify ways to reduce the probability or the impact of the risk
- **Accept** – Nothing will be done

- **Transfer** – Make another party responsible for the risk (buy insurance, outsourcing, etc.)

For each major risk, the team will implement the documented preventive measures and use the stated fallback when the risk occurs. Where the supplied documents do not specify a separate contingency, this remains an item for the Risk Manager and sponsor to agree.

The responses below elaborate the risks in Project Plan Section 8 and other additional risks identified. Their principal approach is mitigation. Scope control avoids unapproved work. No insurance, outsourcing or other transfer arrangement is documented for this project.

1. **Missing or unreliable part data.** Sinha Armaan, supported by Samarth Jethani, will manage incorrect rule results caused by missing values (70%; High impact). Catalogue fields such as thermal resistance may be absent. Keep values optional, require each rule to check its inputs, and use the 40-board regression suite containing valid, invalid and missing-data cases. Missing inputs must produce unchecked results that identify the absent value; engineers may upload a datasheet to fill the gap.
2. **Normalisation drift.** Sparsh Jain will manage inconsistent normalisation of equivalent component inputs (20%; High impact). Changes to prompts or parsing behaviour could cause the same recorded part data to produce different physical interpretations or verdict serialisations. Freeze the normalised input format and serialisation rules, and retain the 100-run determinism and provenance tests to detect unintended changes. Normalisation will remain outside the constraint engine so that changes to interpretation do not alter the engine's deterministic rule evaluation.
3. **Illegal or looping repairs.** Jethani Samarth will manage invalid, repeated or non-terminating repair behaviour (20%; High impact). A model response may select an action outside the permitted repair set, arrive too late, repeatedly fail to resolve a slot, or cause the repair process not to terminate. Test the legal action set with adversarial cases and reject illegal or late selections. When this occurs, use the least-disruptive permitted candidate. Limit repairs to three attempts per slot, enforce the SRS added-slot bound, and escalate the case if a valid repair cannot be completed.
4. **Stream gaps / blank or stalled client.** Armaan Sinha, Lai Yi and Jerral Chua will manage dropped or stalled SSE connections (30%; High impact). Number events so the browser can reconnect from the last event received. The frontend must handle failed or stalled streams instead of displaying an empty page. If recovery fails, display an explicit error and retry control. Stream recovery and event ordering are covered by integration and browser tests.
5. **Stale normalisation-cache interpretation.** Mohika Kataruka will manage the risk of cached component interpretations becoming outdated after the normalisation prompt changes (15%; Moderate impact). Cache entries will be keyed by both the manufacturer part number (MPN) and the normalisation-prompt fingerprint so that a changed prompt forces the component to be parsed again. Stock status, lifecycle information and datasheet links will remain live for each run. Recorded test fixtures will not be used as the live runtime cache.
6. **Supplier rate limits or outages** Armaan Sinha, supported by Jethani Samarth, will manage component-data access risks (40%; High impact). JLCPCB or Mouser restrictions can prevent sourcing and push runs beyond the 130-second target.

Apply tight numerical search limits and use saved distributor responses for testing. When a part cannot be found, leave the slot unresolved and show the error rather than inventing a component or reporting success.

7. **Hosting sleep or outage.** Jerral Chua, supported by Armaan Sinha, will manage hosting and demo continuity (30%; Moderate impact). Free hosting may sleep when idle or fail during a demonstration. Use health checks, monitoring and the deployment runbook, and maintain a working local setup. If hosting fails, switch to the local setup. Keep the uptime record for the hosted system separate from any local demonstration.
8. **Insufficient usability participants.** Faye Chan will manage recruitment of ten first-time users (25%; Moderate impact). Without ten participants, the required usability result cannot be demonstrated at M5. Recruit the participants before M5 and review recruitment progress at team meetings. The source plan does not specify a replacement-participant or reduced-sample contingency; any unresolved shortfall must be raised with the Project Manager before validation review.
9. **Insufficient seeded analytics data.** Lai Yi and Faye Chan will coordinate analytics readiness and validation (10%; Moderate impact, reduced from 30%). Without completed sample runs, the statistics pages may be empty at the demonstration. Prepare at least twenty sample runs before M4 and verify the statistics views as part of validation. Status, 7 October: mitigated. tools/seed_analytics.py writes 25 completed runs across five projects. The runs are generated events, but every figure is computed by the real recorder, store and analytics code, so no statistic is entered by hand. Live runs are added as they happen.
10. **Slow model responses.** Armaan Sinha and Mohika Kataruka will manage model latency and repeated normalization (40%; Moderate impact). Live runs currently take 50 to 130 seconds, at the edge of the DS-5 limit of 130 seconds. Cache normalized part data and look up likely parts early. When the model is too slow to select a repair, use the cheapest permitted fix. The constraint engine must check the resulting board; the model does not make pass/fail decisions.
11. **Scope changes.** Mohika Kataruka will control additions to the agreed scope (25%; High impact). Record the effect of each request on effort and delivery dates and obtain written sponsor approval. The current SRS already includes memory browsing, the walkthrough and standalone BOM validation; further changes to these features must follow the same process. Schematic capture, PCB layout and routing, simulation and multi-server scaling remain out of scope.
12. **Schedule delay.** Mohika Kataruka, supported by the technical work-package owners, will monitor dependency delays (35%; High impact). The contract freeze (M2) was 23 September; a late contract or engine delays downstream integration. Review progress weekly and verify milestone evidence. Because the prototype is already deployed, the two-week uptime period will begin by 21 October. Missed evidence must be reported at review rather than treating an incomplete phase as closed.
13. **Effort underestimated.** Mohika Kataruka will compare actual effort with estimates at each milestone (40%; Moderate impact). The Project Plan identifies a 12.9% excess over the 1,106-hour budget. Re-estimate the remaining work at every milestone. The documented response is to use contingency first and then agree to any scope reduction with the sponsor. The 35.7-hour contingency is already included in the total estimate and is not extra capacity.

14. **Knowledge concentration. Sparsh Jain, supported by Sinha Armaan, will manage the risk that one developer holds most knowledge of the codebase (40%; High impact). The source was imported in a single commit and the analytics work was built by one person. Every change gets a non-author review, each module has a second named reviewer, and design decisions are recorded in the repository. Contingency: a handover session for any module whose owner becomes unavailable.**
15. **Repository access. Jerral Chua will manage the risk that the campus GitLab is unreachable off the NTU network (50%; Low impact). Work is committed locally and pushed on the NTU network or VPN. Every member keeps an up-to-date clone before each milestone.**
16. **Model provider outage or credit exhaustion. Sinha Armaan will manage loss of the language model (25%; High impact). Credit use is checked weekly against the $200 allocation. The recorded walkthrough and the seeded analytics corpus let the prototype be demonstrated without the model. The engine never depends on the model for a verdict.**
17. **Team availability. Mohika Kataruka will manage reduced availability during recess and examination weeks (40%; Moderate impact). Availability is collected at the start of each phase and tasks are rebalanced at the weekly meeting.**

2.5 RISK MONITORING, CONTROLLING, AND REPORTING

Risk status will be reviewed at the weekly team meeting and each milestone. Owners will report changes in likelihood or impact, progress on mitigation, observed failures and supporting test evidence. Faye and Samarth coordinate the write-up and Mohika keeps the risk register current. QA reports open defects weekly. (Project Plan, Sections 2, 8 and 10.)

A “Top 10 Risk List” will be maintained by the project team and will be reported as a component of the project status reporting process for this project. RED risks are reviewed first, followed by YELLOW and GREEN risks. The list will be updated when new risks are identified or existing risks change.

Every change request will be assessed for effects on scope, effort, dates, dependencies and quality evidence. Mohika will obtain the sponsor’s written approval before an agreed scope change is incorporated into the controlled project records.

Serious problems including a wrong pass result or a security issue go directly to Mohika. A phase remains open until the required evidence is available and the sponsor agrees. No completion of these controls is claimed by this plan.

3 TOOLS AND PRACTICES

The Risk Log will be maintained by Mohika and reviewed as a standing item at team meetings. GitLab issues and the Trello board track actions, owners, defects and milestones. The campus GitLab repository records reviewed changes and release tags. pytest and Playwright supply verification evidence; coverage on 7 October is 91% for the backend and 97% for the constraint engine. A GitLab CI pipeline that blocks merges below 80% backend or 95% engine coverage is due on 21 October (Jerral). Release requires evidence for each design target and no open defects causing a wrong pass, blank page or serious security issue. (Project Plan, Sections 2, 9 and 10.)

RISK MANAGEMENT PLAN APPROVAL

The undersigned acknowledge they have reviewed the **Risk Management Plan** for the FUZE project. Changes to this Risk Management Plan will be coordinated with and approved by the undersigned or their designated representatives.

Signature:		Date:	6 Oct 2026
Print Name:	Mohika Kataruka		
Title:	Project Manager		
Role:	Risk Manager		

Signature:		Date:	6 Oct 2026
Print Name:	Jethani Samarth		
Title:	QA Manager		
Role:	Risk Plan Preparation Lead		

Signature:		Date:	6 Oct 2026
Print Name:	Faye Chan		
Title:	QA Engineer		
Role:	Risk Plan Preparation Lead		

Signature:		Date:	
Print Name:	Ajith Santhisenan		
Title:	Project Sponsor		
Role:			

APPENDIX A: REFERENCES

The following table summarizes the documents referenced in this document. Locations are relative to the SC3040 project folder, except the separately supplied template.

Document Name and Version	Description	Location
FUZE Project Plan Version 1.1 7 Oct 2026	Sections 2, 5.2, 8–10: roles, estimates, initial risks, responses and project controls.	Lab 3/ FUZE_Project_Plan.docx
System Requirement Specifications Revision 1.0 14 Sep 2026	Functional requirements, sixteen use cases, constraint rules, missing-data handling and open issues.	Lab 2/ Final_System Requirement Specification.docx
FUZE Software Quality Assurance Plan Revision 3.0 15 Sep 2026	Verification, validation, coverage targets, evidence and review practices.	Lab 2/ Final_FUZE_Quality_Plan.docx
Meeting Minutes 17 Sep 2026	Core risk themes and technical responsibilities.	Lab 3/ 17th September - Meeting Minutes.docx
CDC UP Risk Management Plan Template 30 Nov 2006	Supplied structure, probability bands, impact matrix, response categories and approval format.	Separate upload: CDC_UP_Risk_Management_Plan_Template.docx

APPENDIX B: KEY TERMS

The following table provides definitions for terms relevant to the Risk Management Plan.

Term	Definition
Risk	An uncertain event or condition that, if it occurs, may affect one or more project objectives.
Risk Owner	The team member assigned responsibility for monitoring a risk, watching for its trigger conditions and coordinating the agreed response.
Risk Register / Risk Management Log	The controlled project record contains identified risks and, where available, their owners, likelihood, impact, triggers, mitigation, contingency and review status.
Likelihood	An assessment of how likely a risk is to occur. A likelihood remains unassessed where the project does not have sufficient evidence to support a rating.
Impact	The consequence a risk could have on project objectives such as correctness, performance, schedule, availability, quality or milestone completion.
Trigger	An observable event, condition or threshold indicating that a risk may be developing or has occurred.
Mitigation	An action taken before a risk occurs to reduce its likelihood, reduce its impact, or both.
Contingency	A planned fallback action to be taken if a risk materialises or a mitigation is unsuccessful.
Residual Risk	The risk remaining after mitigation or other controls have been applied. No residual rating is assumed unless it has been specifically assessed.
Escalation	Raising a risk or issue to the appropriate project authority when it requires prompt attention, a decision or action beyond the risk owner's responsibility.
Scope Change	A proposed alteration to the agreed project scope that requires impact assessment and, for this project, written sponsor approval before incorporation.

Provenance	Evidence identifying the origin or source of a data value so that it can be traced and evaluated.
Normalisation	Conversion of source component information into a consistent internal representation before it is evaluated by the constraint engine.
Normalisation Drift	A condition in which equivalent inputs are normalised differently over time, potentially producing inconsistent interpretations or outputs.
Determinism	The property that the same controlled inputs produce the same interpretation and output when the relevant processing conditions are unchanged.
MPN	Manufacturer Part Number; the identifier assigned by a component manufacturer to a particular component or product.
Prompt Fingerprint	An identifier representing the normalisation prompt or its version, used with the MPN to prevent cached results from being reused after the prompt changes.
SSE	Server-Sent Events; the event-stream mechanism used by the application to send design-run progress from the server to the browser.
Unchecked	A rule result indicating that required information is unavailable or insufficient to evaluate the rule. An unchecked result must not be represented as a pass.
Fixture	A recorded and controlled test input or external-service response used to make testing repeatable. A fixture is not treated as live supplier or runtime data.